Optimus - Tutorial 1

Nicholas Andre G. Johnson and Aleksandr B. Sahakyan

16 August 2018

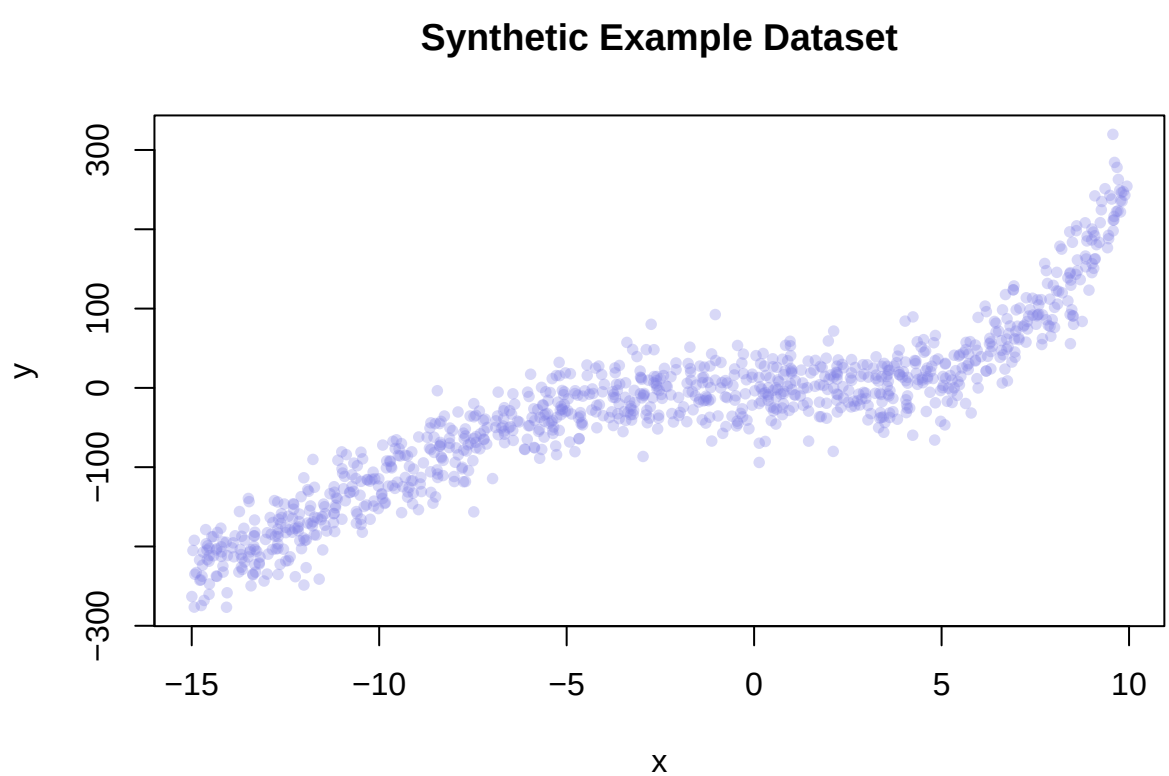
Example 1: Finding coefficients for a polynomial function

In this example, we shall use Optimus to find the coefficients of the polynomial function that is known to represent the observations y the best. This, of course, is a simple task that can be addressed more robustly by least-squares linear model fitting. However, starting with this example will focus our attention on the organisation of the Optimus input, rather than the complexity of the task.

First of all, let us create some data for the example.

```
set.seed(845)
x <- runif(1000, min = -15, max = 10)
y <- -1 * x - 0.3 * x^2 + 0.2 * x^3 + 0.01 * x^4 + rnorm(length(x), mean = 0,
    sd = 30)
```

The good side of this noisy data generation is that we know the original function that describes it: $y = -1.0x - 0.3x^2 + 0.2x^3 + 0.01x^4$. Hence, we can check how well Optimus performs at finding the correct coefficients. The synthetic "real world" noisy data that we generated looks like this:

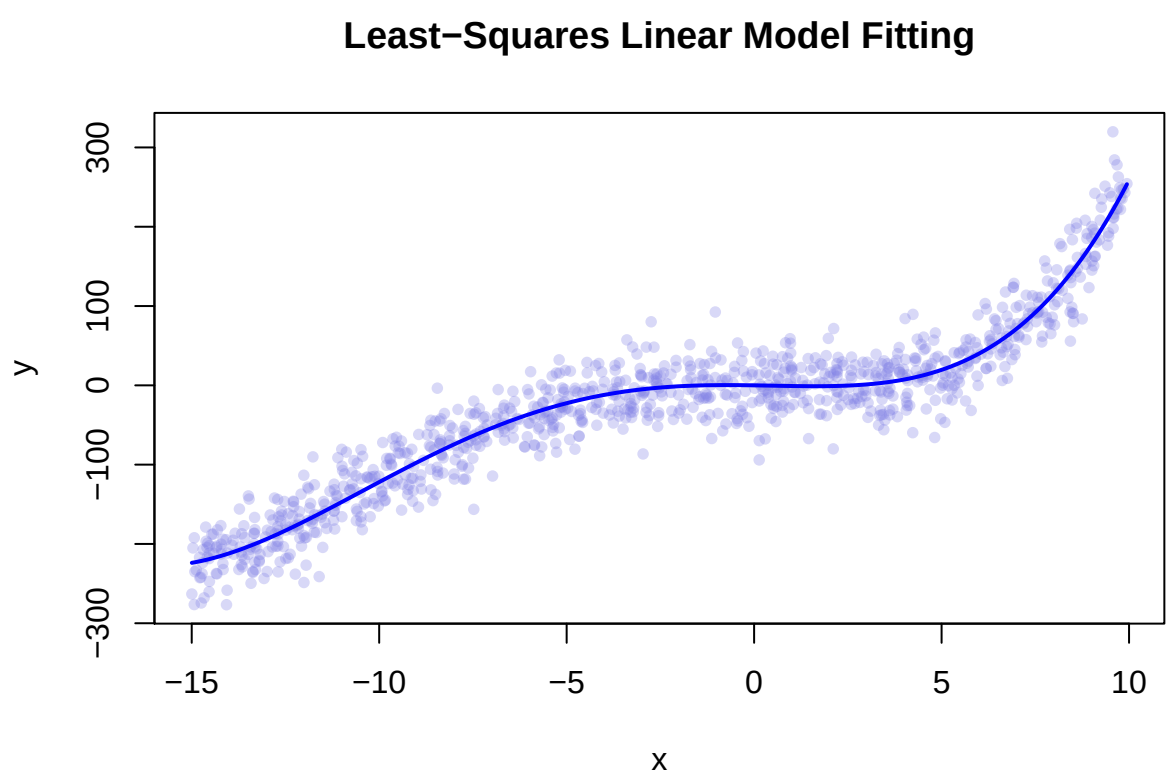


Before we turn to Optimus, let us see how the proper linear model fitting will perform using this data.

```r
lm.model <- lm(y ~ x + I(x^2) + I(x^3) + I(x^4) + 0)
lm.model
```

```
##
## Call:
## lm(formula = y ~ x + I(x^2) + I(x^3) + I(x^4) + 0)
##
## Coefficients:
##        x    I(x^2)    I(x^3)    I(x^4)
## -0.74056  -0.30735   0.19777   0.00991
```

The least-squares linear model fitting for the coefficients to the known functional form is quite close to the original equation: $y = -0.741x - 0.307x^2 + 0.198x^3 + 0.010x^4$.



The RMSD between the observed data y and the linear model fitting outcome is:

```r
y.pred <- predict(newdata = data.frame(x = x), object = lm.model)
sqrt(mean((y - y.pred)^2))
```

```
## [1] 28.82655
```

which is even slightly better in describing the noisy data, as compared to the maximum possible RMSD based on the de-noised data:

```r
y.realdep <- -1 * x - 0.3 * x^2 + 0.2 * x^3 + 0.01 * x^4
sqrt(mean((y - y.realdep)^2))
```

```
## [1] 28.85858
```

Defining Optimus Inputs

Now we can set up the inputs for the Optimus run. We shall use the model $k_1x + k_2x^2 + k_3x^3 + k_4x^4$ to fit the y observables based on the values for x. The dependent functions that are needed for setting up an Optimus run, are given as inputs in the Optimus function call.

First, we need to create an object K, which stores the initial values for the parameter(s) to be optimised. K can be an object of any type. From a single numeric or character value to a vector of values or a data frame holding, say, Cartesian coordinates of a molecule to be optimised. The only requirement for K is that it should be something alterable (via a rule function $r()$, see below) and something that influences the outcome of another required model function - $m()$ (see below). In this example, we have 4 coefficients to optimise from some random initial state. We can thus make K be a numeric vector of size 4. Let us start from all the components being 1.0, which, as entries in K, can be both named and unnamed. Though not the case here, the entry-named data for K can be essential for some models that specifically use coefficient names, for instance when a system of ODEs is used in the model function $m()$.

```
K <- c(k1 = 1, k2 = 1, k3 = 1, k4 = 1)  # entries are named as k1, k2, k3 and k4
```

Second, we should create the function m for the model. The function m should be designed to operate on the whole set of parameter snapshot K and return the corresponding observable object O. Please note that the size of K and O are not necessarily to match, depending on the nature of the model used. Operating on K is the only hard condition on $m()$, which can optionally operate on other data as well. In our situation, the function $m()$ should operate on the provided instance of four coefficients (in the object K), and, additionally on the values x. It should then return a vector of observations O (to be compared with y target observations) of the same size as vector x. Any additional data required by the model, in our case an object with the set of 1000 x values, must be provided to the function in an input variable DATA, a list holding the additional data that must be accessed by $m()$ and $u()$ (see below). The variable DATA must be provided to Optimus in the function call, and $m()$ must take it as an input (in the case that neither $m()$ nor $u()$ require additional data, the two functions should still be created such that they take a variable DATA as an input, and the variable DATA passed to Optimus will be set to NULL).

```
DATA <- NULL
DATA$x <- x
DATA$y <- y

m <- function(K, DATA) {
    x <- DATA$x
    O <- K["k1"] * x + K["k2"] * x^2 + K["k3"] * x^3 + K["k4"] * x^4
    return(O)
}
```

At this point, calling $m(K = K, DATA = DATA)$ will return the predicted O set from the initial, non-optimal values for K, hence rather far from the target $O^{trg} = y$.

In this example, the optimisation goal is for the O model outcomes to come as close as possible to the target observations y, to be achieved by optimising the coefficients K. The object y holding the target values therefore also needs to be specified and given as an input to the $Optimus()$ function (as an entry in the DATA variable), just like x required, in this example, by the function $m()$.

Now, we need to define how the performance of a given snapshot of coefficients K is to be evaluated. For Optimus, this is done by specifying a function $u()$, which should necessarily take as inputs O (the output of $m()$) and the variable DATA. The output should have two components, Q holding a single number of the quality of the K coefficients, and E holding a (pseudo)energy for the given snapshot K. It is important that the returned (pseudo)energy value is lower for better performance/version of K, never vice-versa. The Q component of the $u()$ function output is only used for plotting the optimisation process, and, if desired, can just repeat the value of the E component.

For our example, the $u()$ function will assess the agreement between the snapshot of predictions O and the complete set of real observables (target) y. Here, we can use RMSD between O and y as a measure of K snapshot quality (Q). Since bettter agreement means better RMSD, it can be directly used as a pseudo-energy (E), without putting a negative sign or performing some other mathematical operation on Q.

```
u <- function(O, DATA) {
    y <- DATA$y
    Q <- sqrt(mean((O - y)^2))
    E <- Q  # For RMSD, <-> negative sign or other mathematical operation
    # is not needed.

    RESULT <- NULL
    RESULT$Q <- Q
    RESULT$E <- E
    return(RESULT)
}
```

And finally, we need to define the rule, by which the K coefficient vector is to be altered from one step to another. This is done by defining a rule function $r()$ that must take K, and return an object equivalent to K, but with some alteration(s). In this example, for each snapshot of K, we shall randomly select one of its four coefficients, then either increment or decrement (chosen randomly) it by 0.0005, returning the altered set of coefficients.

```
r <- function(K) {
    K.new <- K
    # Randomly selecting a coefficient to alter:
    K.ind.toalter <- sample(size = 1, x = 1:length(K.new))
    # Creating a potentially new set of coefficients where one entry is altered
    # by either +move.step or -move.step, also randomly selected:
    move.step <- 5e-04
    K.new[K.ind.toalter] <- K.new[K.ind.toalter] + sample(size = 1, x = c(-move.step,
        move.step))

    ## Setting the negative coefficients to 0 (not necessary in this example,
    ## useful for optimising rate constants): neg.ind <- which(K.new < 0)
    ## if(length(neg.ind)>0){ K.new[neg.ind] <- 0 }

    return(K.new)
}
```

All the constructed objects (K) and functions (m, u, r), as well as the data required by $m()$ and $u()$ (stored in the variable DATA) should be defined in an R session and given to Optimus as inputs. The users are free to define some dependencies as additional files (for example initial protein geometry for a Monte-Carlo optimisation), which should be called from within the function definitions.

Acceptance Ratio Annealing Optimus Run

Having constructed K, DATA, $m()$, $u()$ and $r()$, we are now ready to call Optimus. Let us first investigate the Acceptance Ratio Annealing (SA) version of Optimus on 4 processors (the vast majority of personal computers currently have 4 processors), which can be executed as follows:

```
Optimus(NCPU = 4, K.INITIAL = K, rDEF = r, mDEF = m, uDEF = u, OPT.TYPE = "SA",
    DATA = DATA, OPTNAME = "poly_4_SA", LONG = FALSE)
```

Note that the field $LONG = FALSE$ is included in the function call so that all data from the optimsation process is saved. Calling Optimus with $LONG = TRUE$ will result in a memory saving optimisation process

(more details in the Advanced User Manual). Of the 4 optimisation replicas, the second and fourth processors found the best parameter configuration (lowest RMSD):

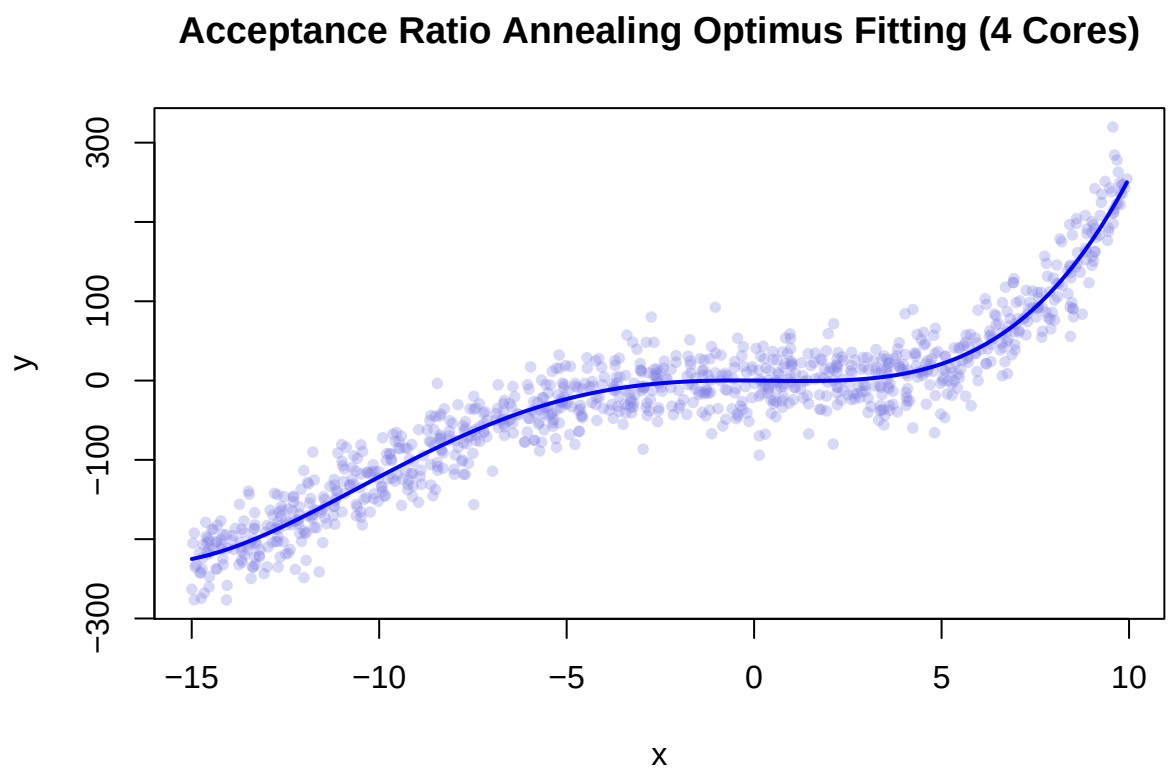


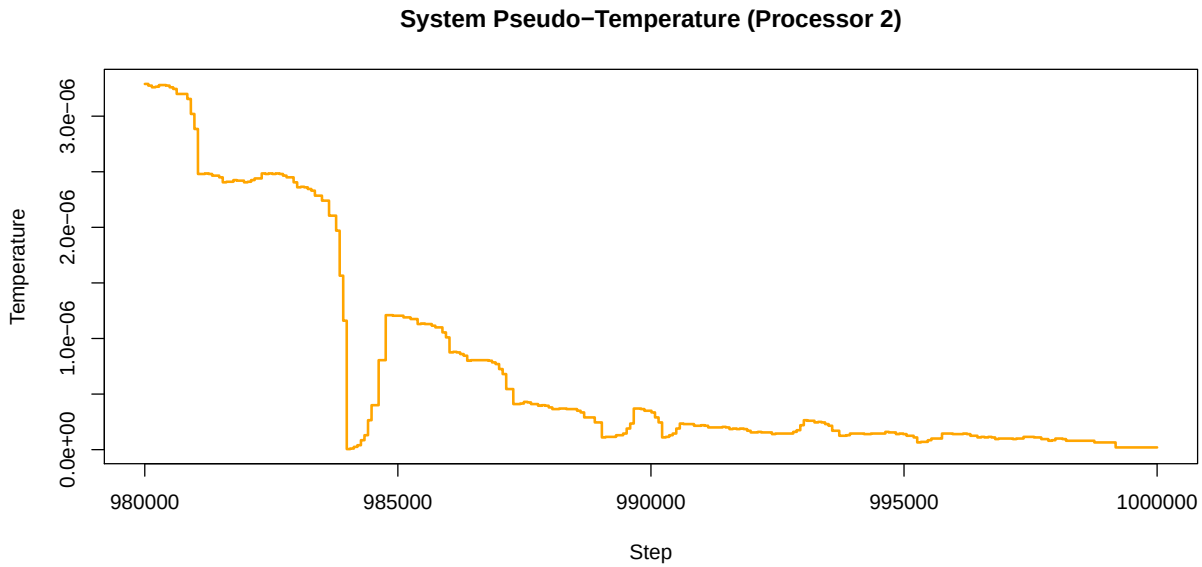
Table 1: 4 Core Acceptance Ratio Annealing Optimus Run Results

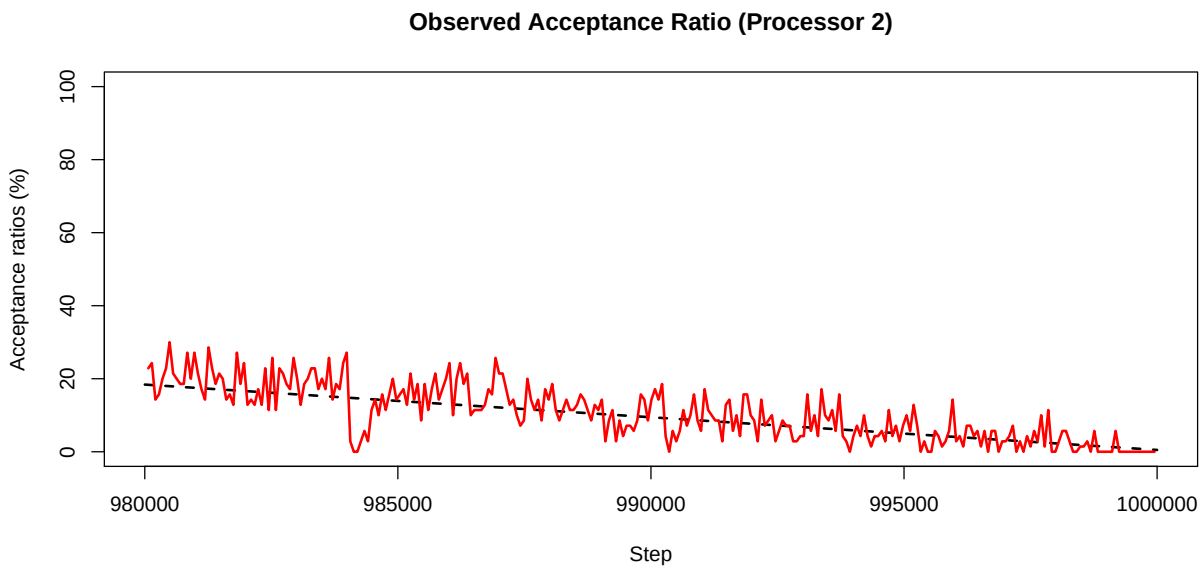
	E (RMSD)	K1	K2	K3	K4
Processor 1	28.857	-0.1560	-0.2850	0.1905	0.0095
Processor 2	28.841	-0.3760	-0.2825	0.1920	0.0095
Processor 3	28.864	-0.1045	-0.2820	0.1905	0.0095
Processor 4	28.841	-0.3760	-0.2825	0.1920	0.0095

The equation recovered by Processor 2 (and 4) is $y = -0.3760x - 0.2825x^2 + 0.192x^3 + 0.0095x^4$.

Notice that although the RMSD of this solution, 28.841, is greater than the RMSD of the least squares solution, 28.82655, it is less than the RMSD of the de-noised data found above, 28.85858.

The two graphs below illustrate the evolution of the system pseudo-temperature in response to alterations made by the Temperature Control Unit (TCU) as a function of the optimsation iteration and the observed acceptance ratio as a function of the optimisation iteration respectively. The graphs show data from the last 20 000 iterations of the optimisation executed by Processor 2.





In the graph titled "Observed Acceptance Ratio (Processor 2)," the solid red line tracks the observed acceptance ratios calculated by Optimus at the end of each STATWINDOW and the dashed black line tracks the target acceptance ratio based on the annealing schedule. From the above two graphs, notice that while the observed acceptance ratio tracks the target acceptance ratio closely, the system pseudo-temperature changes significantly and non-monotonically. This illustrates that the adaptive thermoregulation of the TCU allows Optimus to effectively anneal the system acceptance ratio.

Replica Exchange Optimus Run

Let us now consider the Replica Exchange version of Optimus on 12 processors. Let us reiterate that the purpose here is to illustrate how to run an optimisation using the Replica Exchange version of Optimus; this method is of course not the most robust method to solve this simple task and is likely overkill.

In addition to the arguments specified above, the Replica Exchange version of Optimus also requires an input

variable ACCRATIO which is a vector that defines the acceptance ratio to be used for each of the Replicas initiated, 12 in this case (note that the length of ACCRATIO must always be equal to the argument NCPU).

```
ACCRATIO <- c(90, 82, 74, 66, 58, 50, 42, 34, 26, 18, 10, 2)
```

Having defined the acceptance ratios for each level, the optimisation can be executed as follows:

```
Optimus(NCPU = 12, K.INITIAL = K, rDEF = r, mDEF = m, uDEF = u, ACCRATIO = ACCRATIO,
    OPT.TYPE = "RE", DATA = DATA, OPTNAME = "poly_12_RE", LONG = FALSE)
```

Of the 12 optimisation replicas, replica 8 finds the best parameter configuration (lowest RMSD):

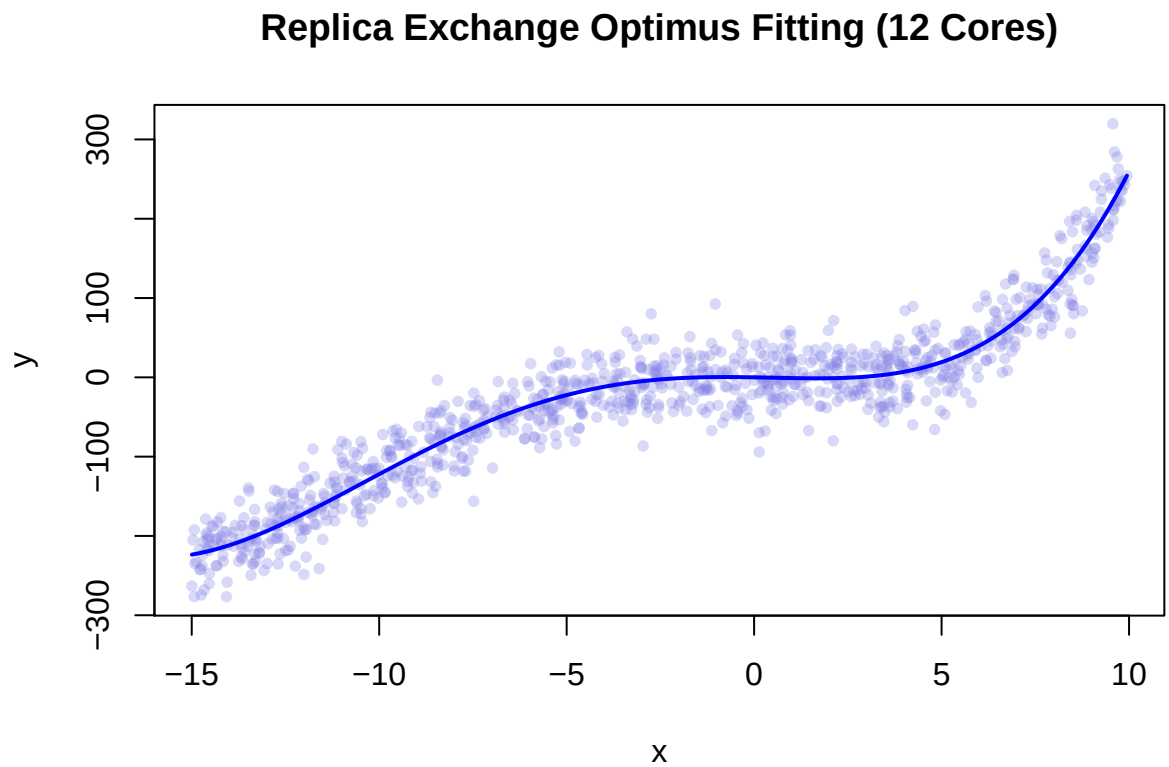


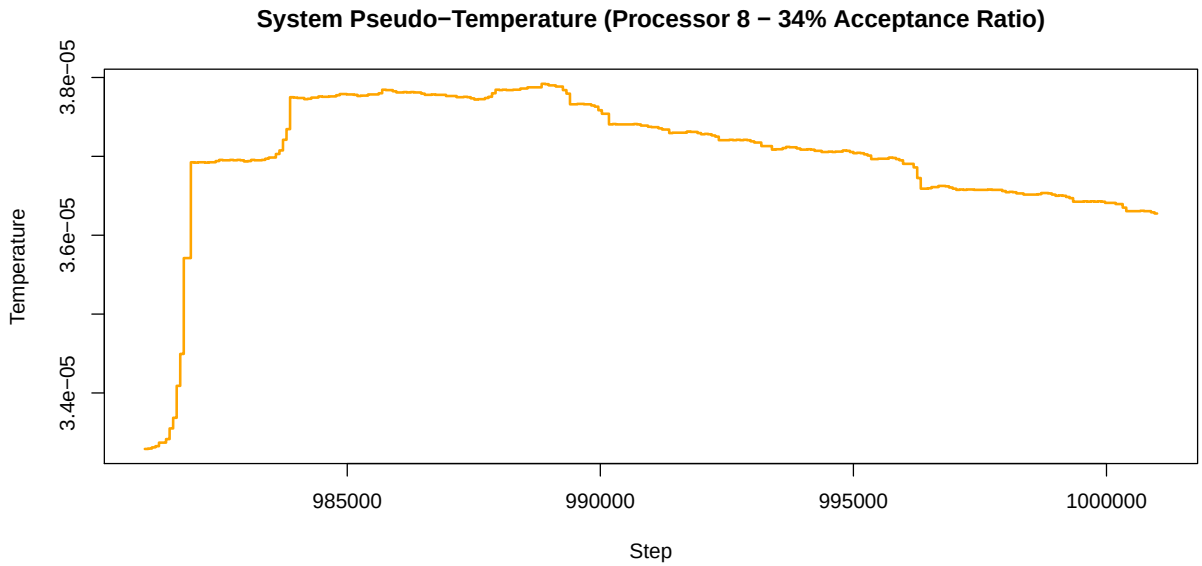
Table 2: 12 Core Replica Exchange Optimus Run Results

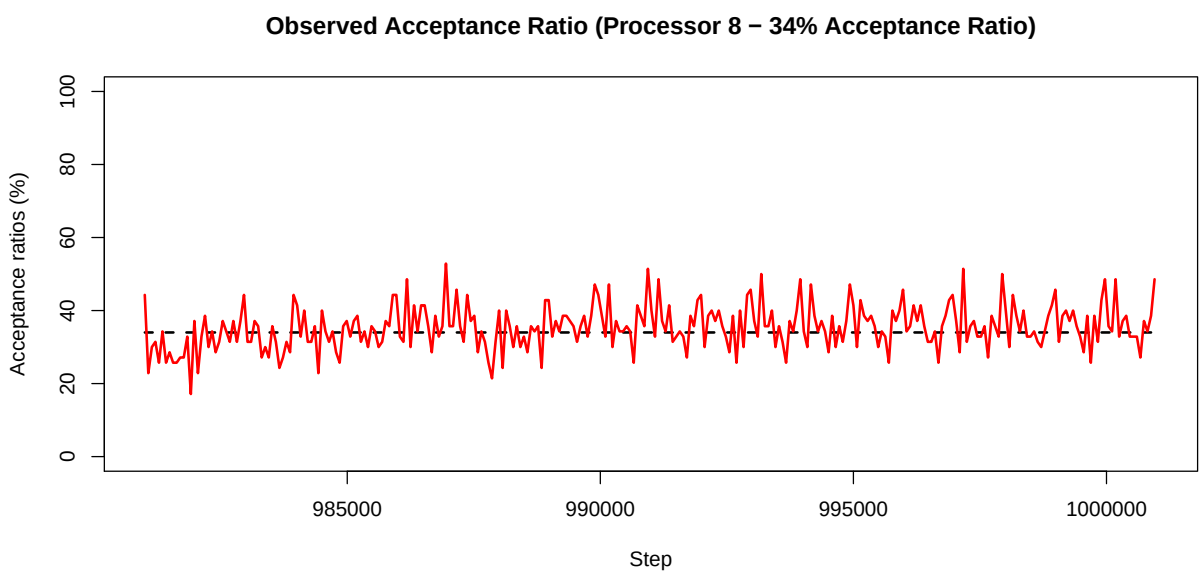
	Replica Acceptance Ratio	E (RMSD)	K1	K2	K3	K4
Processor 1	90	29.71934	-3.2760	-0.5760	0.2395	0.0135
Processor 2	82	29.51819	-3.4445	-0.3755	0.2300	0.0115
Processor 3	74	28.88707	-0.0340	-0.2375	0.1870	0.0090
Processor 4	66	30.10499	-3.8095	-0.6630	0.2435	0.0140
Processor 5	58	29.06293	-2.3575	-0.4180	0.2200	0.0115
Processor 6	50	29.70906	-3.7360	-0.3785	0.2320	0.0115
Processor 7	42	28.85095	-1.1370	-0.3515	0.2045	0.0105
Processor 8	34	28.82721	-0.8175	-0.3130	0.1990	0.0100
Processor 9	26	28.84057	-0.5760	-0.2740	0.1940	0.0095
Processor 10	18	29.48785	-2.7805	-0.5420	0.2325	0.0130
Processor 11	10	28.85095	-1.1370	-0.3515	0.2045	0.0105
Processor 12	2	29.41377	1.2255	-0.0895	0.1645	0.0070

The equation recovered by Processor 8 is $y = -0.8175x - 0.313x^2 + 0.199x^3 + 0.01x^4$.

Notice that the RMSD of this solution, 28.82721, is less than the RMSD of the Acceptance Ratio Annealing solution, 28.841, and only slightly greater than the RMSD of the least squares solution, 28.82655.

Let us now briefly examine the evolution of the system pseudo-temperature in response to alterations made by the Temperature Control Unit (TCU) as a function of the optimsation iteration and the observed acceptance ratio as a function of the optimisation iteration for the Replica Exchange version of Optimus. The following two graphs represent data from the last 20 000 iterations of the optimisation replica running on Processor 8 (34% target acceptance ratio).





Notice that in the observed acceptance ratio graph, the dashed line indicating the target acceptance ratio is constant (as opposed to linearly changing as in acceptance ratio annealing). This is because each processor in the replica exchnage mode has a single target acceptance ratio, as articulated in the previous section of this manual which addressed the theoretical underpinnings of Optimus. Here again, adaptive decisions

made by the TCU result in non-monotonic, non uniform psuedo-temperature adjustments while the observed acceptance ratios fluctuates relatively closely around the target ratio.

Summary

We now understand the input requirements to interface with the Acceptance Ratio Annealing and Replica Exchange versions of Optimus. In this example, both the Acceptance Ratio Annealing and Replica Exchange Optimus versions retrieved solutions having a lower RMSD than the de-noised data and only a slightly greater RMSD than the optimal Least Squares solution. Replica Exchange resulted in a better solution than Acceptance Ratio Annealing, at the cost of greater computing resources.

Table 3: Summary of Solutions

	E (RMSD)	K1	K2	K3	K4
De-noised Function	28.85858	-1.00000	-0.30000	0.20000	0.01000
Optimus (Acceptance Ratio Annealing)	28.84100	-0.37600	-0.28250	0.19200	0.00950
Optimus (Replica Exchange)	28.82721	-0.81750	-0.31300	0.19900	0.01000
Least Squares	28.82655	-0.74056	-0.30735	0.19777	0.00991